

- (d) An event in the probability that will never be happened is called as
 a. Unsure event
 b. Sure event
 c. Possible event (1 Mark)
 d. Impossible event
- (e) Which of the following is a discrete random variable?
 (a) The time it takes for a plant to grow.
 (b) The number of errors on a webpage.
 (c) The height of a building. (1 Mark)
 (d) The temperature of a room.

Question 2 (10 Marks)

1. The probability of the selecting person X is $\frac{2}{3}$, and Y is $\frac{3}{5}$ in the job application. What is the probability that both of them are selected? (2 Marks)
2. A password is 6 characters long, using 26 letters and 10 digits, no repetition. How many passwords? (2 Marks)
3. Consider the discrete random variable X with the following distribution

X	0	1	2	3
$P(X = x)$	0.1	0.3	0.4	0.2

Calculate the following:

- (a) the expected value of X , $E[X]$. (2 Mark)
- (b) the expected value of $3X + 2$, $E[3X + 2]$ (1 Mark)
- (c) the variance, $Var[X]$ (2 Marks)
- (d) the standard deviation, σ_X . (1 Mark)

Question 3 (10 Marks)

1. In a box, there are 3 orange, 5 white, and 2 blue balls. If a ball is picked up randomly, what is the probability that it is neither orange nor a white ball? (2 Marks)
2. If ${}_nC_4 = {}nC_{10}$, find the possible value(s) of n . (2 Marks)
3. A continuous random variable X has a probability density function

$$f(x) = \begin{cases} kx^2 & 0 < x < 2 \\ 0 & \text{elsewhere} \end{cases}$$

Determine the following:

- (a) the value of k , (2 Marks)
- (b) the expected value, $E[X]$, (2 Marks)
- (c) the variance, $Var[X]$. (2 Marks)



Faculty: Information Technology
Academic Year: 2025-2026, Semester: 1
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MODULE NAME: Probability and Statistics
Module Code: STAT 8225
Lecturer: Jean Paul NSABIMANA
Date of the Exam: 21st October 2025
Group: Day (A and D). Exam Duration: 2 Hours

EXAM INSTRUCTIONS:

1. The Paper is made up of **three** questions.
2. Total Marks of the Exam: **30 Marks**.
3. Phones and Electronic Devices are strictly forbidden unless instructed otherwise.
4. Scientific calculators are allowed but not to be shared in the examination room.

Question 1 (10 Marks)

1. True/False questions.

- (a) An event is an element of the sample space. **1Mark**
- (b) If E and F are mutually exclusive events, $E \cap F$ is impossible event. **1Mark**
- (c) In discrete probability distribution, the sum of probabilities must equal 1. **1Mark**
- (d) An event and its complement can occur at the same time. **1 Mark**
- (e) Probability of an event can be greater than 1 or less than 0. **1 Mark**

2. Multiple choice questions.

- (a) The probability of an event that is certain to occur is:
a. Between 0 and 1, b. 0.5 c. 0, d. 1. **(1 Mark)**
- (b) The probability that a hard drive fails in a year is 0.02. What is the probability that it does not fail?
a. 0.02 b. 0.98 c. 1 d. 0.5. **(1 Mark)**
- (c) Given $P(A) = 0.4$, $P(B) = 0.5$, $P(A \cap B) = 0.2$, what is the $P(A|B)$?
a. 0.1 b. 0.2 c. 0.4 d. 0.6. **(1 Mark)**